Compass MCP - BETA

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Compass MCP — Product Overview

> **What is Compass MCP?**
> Compass is a Model Context Protocol (MCP) server that connects AI assistants (Claude, Cursor, etc.) to your data infrastructure. It unifies **DataHub** (data catalog), **Superset** (visualization), and **Trino** (SQL engine) into a single conversational interface — letting anyone ask questions about data in plain English.

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HERO FEATURE 1

1. Domain-Specific Analytics Skills — The Brain

> **Skills are the single most powerful feature of Compass.** They encode deep knowledge about Groww's data — which tables to use, how to filter, what columns mean, and which queries are safe to run. Users don't need to know the schema. They just ask questions.

The Problem

Groww's data platform has **hundreds of tables** across multiple schemas, with **billions of rows**, inconsistent naming, and complex join relationships. Even experienced analysts spend significant time finding the right table, understanding column semantics, and writing efficient queries.

The Solution

Each skill is a curated knowledge package containing:

What the Skill Knows	Example
Which tables to use	"For stock orders, use `core_stocks_order`, NOT the stale `stocks_order`"
Partition columns	"Always filter on `order_date` — this table has 2.5 billion rows"
Column semantics	"`product = 'CNC'` means delivery, `'MIS'` means intraday"
All filter values	"Order status: `EXECUTED`, `CANCELLED`, `REJECTED`, `FAILED`"
Optimized SQL patterns	Pre-written templates for the 10 most common questions per domain

Data quality warnings	"This table has future-dated rows — always cap your date range"
Stale table alerts	"DO NOT USE `invest_user_fid_info` — stale since Jun 2024"
Join keys	"Universal key: `user_account_id` (format `ACC<digits>`)"

How It Works — End to End

```
User: "What was the stock order success rate yesterday?"

Step 1: Compass detects this is a Stocks & Trading question
Step 2: Loads the Stocks & Trading skill
Step 3: Skill tells the AI:
        - Table: core_stocks_order (not the stale stocks_order!)
        - Partition: order_date (MUST filter — 2.5B rows)
        - Success = order_status = 'EXECUTED'
Step 4: AI writes optimized SQL:
        SELECT order_date,
               COUNT(*) AS total,
               COUNT(*) FILTER (WHERE order_status = 'EXECUTED') AS executed,
               ROUND(100.0 * COUNT(*) FILTER (WHERE order_status = 'EXECUTED') / COUNT(*), 2) AS rate
        FROM platform_iceberg.core_invest.core_stocks_order
        WHERE order_date = current_date - interval '1' day
        GROUP BY order_date
Step 5: Executes on Trino, returns: "Yesterday's success rate was 87.3% across 14.2M orders"
```

Without skills: User opens SQL Lab, searches for the right table, guesses column names, writes SQL, debugs partition errors, and interprets raw results.

With skills: User asks one question and gets the answer in seconds.

All 16 Skills

#	Skill	Domain	Key Tables	Table Sizes	Example Question
1	**Stocks & Trading**	Equities	`core_stocks_order`, `fact_transaction_stocks`	2.5B, 7.1B rows	"CNC vs MIS turnover this week?"
2	**F&O Analytics**	Derivatives	`fno_order_master`, `fno_pnl_master`	3.8B, 1.9B rows	"Options PnL distribution"
3	**Mutual Funds**	MF	`fact_mutual_fund_order_details_enriched`, `view_mf_user_aum_latest`	Millions	"SIP creation trends this month?"
4	**User Master & Growth**	Growth	`growth_user_master_ultimate`, `user_product_txn_master_fact`	117M, 8.3B rows	"Signup → FID conversion by source"
5	**User Events**	Instrumentation	DataHub `userEvent` entities + `ems_raw`	57B+ events	"What events fire on the home screen?"

6	**Onboarding**	Activation	`onb_usercheckpoints_master`, `fact_onboarding_invest`	117M, 45M rows	"Where do users drop off in KYC?"
7	**UPI Payments**	Payments	`groww_upi_onb_user_fact`, `groww_upi_mandate_fact_table`	66.9M, 27.2M rows	"UPI mandate lifecycle analysis"
8	**MTF & Pledge**	Margin	`aay_mtf_book_agg`, `fact_pledge_transaction`	243, 50M rows	"MTF book size and market share"
9	**Commodities**	MCX	`commodity_day_view_seg`, `commodity_order_master`	684K, 71.6M rows	"Gold vs Silver turnover"
10	**915 Trading**	915 App	`userday_details_915`, `mis_central_915_ab`	1.9M, 17 rows	"915 DAU and MTU trends"
11	**Help & Support**	Support	`hns_helpdesktickets_master_trino`, `hns_chatbot_fact_table_v3`	12M, 135K rows	"Chatbot self-service rate"
12	**Engagement**	Comms	`engagement_pn_narad_master`, `engagement_user_session_daily`	250M/day, 8M/day	"Push notification CTR by campaign"
13	**Context Push**	Notifications	Context-aware push tables	Varies	"Contextual push delivery analysis"
14	**User Financial Health**	Wellness	Financial health tables	Varies	"User investment diversification"
15	**Datahub Navigation**	Meta	DataHub entities	N/A	"How do I find dataset lineage?"
16	**MF Analytics Extended**	MF	Extended MF tables	Varies	"Scheme-level NAV performance"

What Each Skill Contains

Every skill is a comprehensive guide with:

- **Tables Overview** — All relevant tables with row counts, freshness status, grain, and partition columns
- **Key Columns Reference** — Every important column with type, description, and sample values
- **Common Filter Values** — Exact enumeration of status codes, product types, platform values, etc.
- **10+ Example Queries** — Production-ready SQL templates for the most common questions
- **Query Strategy Table** — "If user asks X, use table Y with approach Z"
- **Table Relationships** — Join keys and cross-table navigation
- **Data Quality Notes** — Known issues, stale tables, naming inconsistencies, PII warnings
- **Critical Guardrails** — Partition requirements, row count warnings, timeout prevention

Skill Coverage at a Glance

```
                    Compass Analytics Skills Coverage
```

```
Stocks ──────────────────── 3 tables, 12B+ rows, CNC/MIS/MTF/AMO/GTT/SIP/ETF
F&O    ──────────────────── 9 tables, 6B+ rows, Options/Futures/PnL/Power Traders
MF     ──────────────────── 10+ tables, Orders/SIPs/AUM/NAV/Schemes/Engagement
Growth ──────────────────── 11 tables, 117M users, Signup/FID/AUM/Referral/Uninstall
Events ──────────────────── DataHub metadata + Pinot/Iceberg, 57B+ events
Onb    ──────────────────── 9 tables, Full KYC/eSign/Bank/F&O activation funnel
UPI    ──────────────────── 6 tables, Onboarding/Mandates/Recon/Payouts/Bills
Engage ──────────────────── 9 tables, PN/Email/SMS/WhatsApp/DAU/Sessions
MTF    ──────────────────── 9 tables, Book/Market Share/Shortfall/OCO/Pledge
Commod ──────────────────── 6 tables, MCX Gold/Silver/Crude/PnL/Compliance
H&S    ──────────────────── 7 tables, Tickets/Chatbot/Voice/CSAT/Agent Quality
915    ──────────────────── 7 tables, DAU/MTU/Sessions/Onboarding/Segments
```

For the detailed breakdown of each skill, see the companion document: Compass Analytics Skills Deep Dive

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HERO FEATURE 2

2. Trino SQL Querying — The Engine

> **Run SQL queries against Groww's entire data lake — directly from conversation.** No SQL Lab, no browser tabs, no context switching. Ask a question, get results inline with full formatting.

What You Can Do

Capability	Example Prompt	What Happens
Ad-hoc analysis	"How many stock orders were placed yesterday?"	AI writes SQL, executes on Trino, returns formatted results
Cross-table joins	"Show me FID rate by signup cohort and source"	Joins multiple billion-row tables with proper partition filters
Data validation	"Are there null user_account_ids in the order table?"	Runs validation query, reports findings
Trend analysis	"Daily order volume for the last 7 days"	Time-series query with tabular results
Funnel analysis	"Signup → onboarding → FID conversion funnel"	Multi-step funnel using checkpoint timestamps
Cohort analysis	"D7 retention rate by signup month"	Cohort-based retention computation
Open in SQL Lab	"Open this query in SQL Lab"	Generates pre-populated SQL Lab URL
Save queries	"Save this as 'Daily MF Orders'"	Persists in Superset's Saved Queries

How It Works

```
User: "Top 10 stocks by turnover this week"
```

```
Compass:
  1. Loads Stocks & Trading skill
  2. Identifies core_stocks_order as the right table
  3. Adds partition filter (order_date >= this week's Monday)
  4. Writes optimized SQL with proper aggregation
  5. Executes on Trino via run_trino_query
  6. Returns formatted results:

  | Stock          | Orders  | Unique Traders | Turnover (Cr) |
  |----------------|---------|----------------|---------------|
  | RELIANCE       | 482,391 | 156,203        | 8,234.56      |
  | TCS            | 312,847 | 98,412         | 5,671.23      |
  | HDFC BANK      | 289,156 | 87,654         | 4,892.10      |
  | ...            | ...     | ...            | ...           |
```

Two SQL Engines

Engine	Tool	Best For	Limits
Trino (Direct)	`run_trino_query`	Ad-hoc analysis on the data lake	Read-only, 5,000 rows, 600s timeout
Superset SQL	`superset > execute_sql`	Queries on Superset-connected databases	Read-only, 10,000 rows, 300s timeout

Built-in Safety Guardrails

Guardrail	What It Does
Read-only	Only `SELECT` statements allowed — no writes, updates, deletes
Partition enforcement	Skills ensure partition filters on billion-row tables
Timeout protection	Configurable timeouts (default 120s, max 600s) prevent runaway queries
Row limits	Results capped at configurable limits (default 500, max 5,000)
Stale table prevention	Skills warn against querying deprecated tables
Date range capping	Skills add upper-bound date filters to prevent future-date scans

Query Complexity It Handles

Complexity	Example
Simple lookup	“How many users signed up today?”
Aggregation	“Average order value by product type”
Filtered aggregation	“F&O success rate for power traders on Android”
Multi-table join	“User AUM correlated with trading frequency”
Window functions	“Running 7-day average of daily order volume”

Funnel analysis	“Signup → PAN → Bank → eSign → Onboarding drop-off”
Cohort retention	“M0/M1/M3 retention by signup month and source”
PnL computation	“F&O PnL distribution — how many users are profitable?”
Time-series	“Daily, weekly, monthly trend of any metric”

SQL Lab Integration (BETA)

Don’t just get results — seamlessly transition to interactive exploration:

Action	What Happens
Open in SQL Lab	`"Open this query in SQL Lab"` — Generates a URL with pre-filled SQL, database, and schema
Save query	`"Save this query as 'Weekly MF Report'"` — Persists in Superset’s Saved Queries list
List saved queries	`"Show my saved queries"` — Lists all saved queries with SQL text

Real Conversation Examples

Example 1: Quick metric

```
User:  "How many F&O orders were executed yesterday?"
AI:    Runs query → "12.3M F&O orders were executed yesterday,
       with a 78.2% success rate. Options accounted for 89% of volume."
```

Example 2: Deep analysis

```
User:  "Compare MTF turnover across large cap, mid cap, and small cap stocks
        for the last 30 days, broken down by week"
AI:    Writes a query joining aay_mtf_fid_agg with market cap data,
       aggregates by week and market_cap, returns a formatted table with trends.
```

Example 3: Validation

```
User:  "Are there any orders with future dates in core_stocks_order?"
AI:    Runs: SELECT order_date, COUNT(*) FROM ... WHERE order_date > current_date
       → "Yes, 1,247 orders have dates in 2089-2090. These are likely erroneous.
          The skill already caps date ranges to prevent scanning these."
```

HERO FEATURE 3

3. Dashboards & Visualization — The Face

> **Create charts and full dashboards through conversation — no drag-and-drop UI, no Superset navigation.** Go from question to published dashboard in minutes.

The Full Workflow

```
Step 1: ASK          "Show me daily order volume as a bar chart"
                              ↓
Step 2: PREVIEW       Compass generates an interactive chart preview
                      Returns a clickable Superset Explore URL
                              ↓
Step 3: ITERATE       "Change it to a stacked bar, group by product type"
                      "Add a filter for only EXECUTED orders"
                      "Switch to a line chart instead"
                              ↓
Step 4: SAVE          "Save this chart"
                      Chart is permanently saved in Superset
                              ↓
Step 5: DASHBOARD     "Create a dashboard with this chart and 3 others"
                      Dashboard auto-generated with 2-column grid layout
                              ↓
Step 6: SHARE         Returns dashboard URL — ready to share with the team
```

Chart Types

Type	Config Key	Best For	Example
Line	`xy` (kind: `line`)	Trends over time	Daily order volume trend
Bar	`xy` (kind: `bar`)	Comparisons	CNC vs MIS vs MTF turnover
Stacked Bar	`xy` (kind: `bar`, stacked: true)	Part-of-whole over time	Order volume by product stacked
Horizontal Bar	`xy` (kind: `bar`, orientation: `horizontal`)	Rankings	Top 10 stocks by volume
Area	`xy` (kind: `area`)	Cumulative trends	AUM growth over time
Scatter	`xy` (kind: `scatter`)	Correlations	AUM vs revenue scatter
Pie / Donut	`pie`	Proportions	Platform distribution
Table	`table`	Detailed data	Top 50 users by turnover
Interactive Table	`table` (viz: `ag-grid-table`)	Sortable/filterable data	Full order detail grid
Pivot Table	`pivot_table`	Cross-tabulations	Orders by product x month
Mixed Timeseries	`mixed_timeseries`	Dual-axis overlay	Revenue (bars) + Users (line)

Chart Features

Feature	How to Use
Filters	“Only show EXECUTED orders” → `filters: [{column: "order_status", op: "=", value: "EXECUTED"}]`
Group by / Series	“Break down by product type” → `group_by: {name: "product"}`
Time grain	“Aggregate by week” → `time_grain: "P1W"`
Axis formatting	“Format Y-axis as currency” → `y_axis: {format: "$,.2f"}`
Legend	“Put legend on top” → `legend: {position: "top"}`

Stacking	"Stack the bars" → `stacked: true`
Sorting	"Sort by revenue descending" → `sort_by: ["revenue DESC"]`

Dashboard Creation

Capability	Example Prompt	What Happens
Create from charts	"Create a dashboard with charts 101, 102, 103, 104"	Auto-positioned 2-column grid layout, published
Add to existing	"Add this chart to the Trading Overview dashboard"	Chart added to next available position
With tabs	"Add this chart to the 'F&O' tab"	Chart placed in the specified tab
Custom title	"Create a dashboard called 'Weekly MF Review'"	Named dashboard with description
Publish/unpublish	"Unpublish this dashboard"	Controls visibility

Dashboard Management

Action	Example Prompt
View info	"What charts are in dashboard 42?"
List all	"Show me all published dashboards"
Search	"Find dashboards about mutual funds"
Update title	"Rename this dashboard to 'Q4 Trading Review'"
Add description	"Add a description: Weekly review of trading metrics"
Check filters	"What filters are applied to this dashboard?"
Delete	"Delete dashboard 99" (with confirmation)

Real Conversation Examples

Example 1: Quick chart

```
User:  "Show me daily stock order volume for the last 7 days as a bar chart"
AI:    Creates an XY bar chart using core_stocks_order dataset
       Returns: Explore URL + preview
       "Here's the chart: [link]. You can interact with it in Superset."
```

Example 2: Iterative refinement

```
User:  "Show me MF order trends"
AI:    Creates a line chart of daily MF orders
User:  "Break it down by SIP vs Lumpsum"
AI:    Updates with group_by on order_kind
User:  "Make it a stacked area chart"
AI:    Changes kind to 'area', enables stacking
User:  "Add a filter for only completed orders"
AI:    Adds filter: order_status = 'COMPLETED'
User:  "Save it"
AI:    Saves permanently → "Chart saved as 'MF Order Trends by Kind' (ID: 247)"
```

Example 3: Full dashboard build

```
User:  "I need a trading overview dashboard with:
        1. Daily order volume (line)
        2. Product mix (pie)
        3. Top 10 stocks (table)
        4. Success rate trend (line)"

AI:    Creates 4 charts, saves each, then:
       "Dashboard 'Trading Overview' created with 4 charts. [link]"
```

Example 4: Pie chart with donut

```
User:  "Show platform distribution as a donut chart"
AI:    Creates pie chart with donut=true, dimension=order_platform, metric=COUNT
       Labels show key, value, and percentage
```

Example 5: Dual-axis mixed chart

```
User:  "Show me daily revenue as bars and unique users as a line, on the same chart"
AI:    Creates mixed_timeseries with:
       - Primary Y (bars): SUM(charges)
       - Secondary Y (line): COUNT_DISTINCT(user_account_id)
       - Shared X: order_date with P1D grain
```

Key Tools

Tool	Purpose

`generate_explore_link`	**Preferred** — Creates interactive explore URL (no save)
`generate_chart`	Create chart preview, optionally save permanently
`update_chart`	Modify a saved chart's config
`update_chart_preview`	Iterate on unsaved preview
`get_chart_info`	View chart metadata and config
`get_chart_data`	Get raw data behind a chart
`get_chart_preview`	Get chart preview (URL, ASCII, table, or Vega-Lite)
`list_charts`	Search and filter existing charts
`generate_dashboard`	Create dashboard from chart IDs
`add_chart_to_existing_dashboard`	Add chart to existing dashboard
`get_dashboard_info`	Dashboard metadata + filter state
`list_dashboards`	Search and filter dashboards
`update_dashboard`	Modify title, description, publish state

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4. Data Discovery & Catalog

Find any dataset, dashboard, chart, or data entity across the organization using natural language.

Capability	Example Prompt	What Happens
Search datasets	"Find tables related to mutual fund orders"	Searches DataHub for matching datasets with URNs and metadata
Explore table schemas	"What columns does the stocks order table have?"	Returns all fields with types, descriptions, tags, and glossary terms
Get entity details	"Tell me about the user_master table"	Fetches full metadata: properties, ownership, tags, descriptions

Browse database tables	"What tables exist in the core_invest schema?"	Lists all tables/views in a database schema
Find saved queries	"Show me saved queries for MF analytics"	Lists saved SQL queries with labels and SQL text
Search user events	"What events fire on the home screen?"	Searches DataHub's userEvent entities for instrumentation metadata

Key Tools: `search`, `list_schema_fields`, `get_entities`, `get_dataset_queries`, `get_database_schemas`, `get_database_tables`

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5. Metadata Management

Enrich your data catalog by managing descriptions, ownership, tags, glossary terms, and domains — all through conversation.

Capability	Example Prompt	What Happens
Update descriptions	"Add a description to the order_status column"	Updates with safe merge workflow (fetch → merge → review → write)
Manage ownership	"Add Saurabh as technical owner of this dataset"	Adds/removes owners with type (TECHNICAL_OWNER, BUSINESS_OWNER, DATA_STEWARD)
Add tags	"Tag this dataset as 'production-ready'"	Adds DataHub tags to entities or individual columns
Glossary terms	"Link this column to the 'Revenue' glossary term"	Associates standardized business definitions
Set domains	"Assign this dataset to the Growth domain"	Organizes entities into business areas
Superset tags	"Tag this chart as 'weekly-review'"	Creates and applies tags within Superset

Key Tools: `update_description`, `add_owners` / `remove_owners`, `add_tags` / `remove_tags`, `add_glossary_terms` / `remove_glossary_terms`, `set_domains` / `remove_domains`

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6. Data Lineage

Understand how data flows through your systems — trace any dataset upstream to its sources or downstream to its consumers.

Capability	Example Prompt	What Happens

Upstream lineage	“Where does this table get its data from?”	Shows all upstream data sources and transformations
Downstream lineage	“What dashboards and tables depend on this dataset?”	Shows all downstream consumers
Multi-hop traversal	“Show me the full lineage chain, 3 levels deep”	Traces lineage up to N hops

Key Tool: `get_lineage` — Upstream or downstream with configurable depth (`max_hops`)

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7. Reports & Alerts

Schedule automated reports and set up data-driven alerts.

Capability	Example Prompt	What Happens
Schedule reports	“Send this dashboard as a PDF every Monday at 9am”	Creates scheduled report with email/Slack delivery
Data alerts	“Alert me when daily order count drops below 100K”	SQL-based data alert with threshold
List reports	“Show me all active report schedules”	Lists all reports and alerts
Manage reports	“Disable the weekly MF report”	View, update, or delete schedules

Features: Formats (PNG, PDF, CSV, TEXT) | Delivery (Email, Slack, Webhook) | Cron scheduling

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8. User Identity & Access

Capability	Example Prompt	What Happens
Check identity	“Who am I logged in as?”	Returns DataHub display name, username, and URN
Get MCP token	“Give me a fresh MCP token”	Generates auth token + ready-to-paste Cursor config
Instance info	“How many dashboards do we have?”	Counts of dashboards, charts, datasets, databases, users
Health check	“Is Compass MCP running?”	Service status, version, system info

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9. Architecture & Getting Started

Architecture

```
+------------------+     +------------------+     +------------------+
|                  |     |                  |     |                  |
|  Claude / Cursor |---->|   Compass MCP    |---->|     DataHub      |
|  (AI Assistant)  |     |   (MCP Server)   |     |  (Data Catalog)  |
|                  |     |                  |     |                  |
+------------------+     +--------+---------+     +------------------+
                                  |
                    +-------------+-------------+
                    |                           |
           +--------v---------+       +---------v--------+
           |                  |       |                  |
           |     Superset     |       |      Trino       |
           | (Charts & Dash)  |       |   (SQL Engine)   |
           |                  |       |                  |
           +------------------+       +------------------+
```

Supported Clients

Client	Status
Claude Code (CLI)	Supported
Cursor IDE	Supported
Any MCP-compatible client	Supported

Getting Started

Step 1: Get Your MCP Token
Ask Compass: `"Give me an MCP token"` — returns your token and ready-to-paste config.

Step 2: Configure Your Client
Paste the config into your MCP settings file (e.g., `.cursor/mcp.json`).

Step 3: Start Asking Questions
No SQL knowledge required. Just ask in plain English:

- “How many users signed up last week?”
- “Show me a chart of daily order volume”
- “Create a dashboard with MF and Stocks charts”
- “What tables have data about mutual funds?”

Key Differentiators

Feature	Without Compass	With Compass
Analytics	Know which tables, columns, filters to use	AI loads the right skill and handles it
Writing SQL	Open SQL Lab, know the schema, write queries	“How many orders yesterday?”

Creating dashboards	Navigate Superset, create charts one-by-one, build layout	“Create a dashboard with these 4 metrics”
Finding data	Browse DataHub UI, search manually	“Find tables about stock orders”
Understanding lineage	Click through DataHub lineage graph	“Where does this data come from?”
Managing metadata	Edit fields in DataHub UI one-by-one	“Tag this as production-ready and add a description”

Summary of Tools (80+ capabilities)

Category	Count	Examples
Analytics Skills	**16**	**Domain-specific pre-built knowledge across all business verticals**
SQL Execution	**5**	**Trino queries, Superset SQL, SQL Lab, save queries**
Charts	**8**	**Create, preview, update, get data, get info, list, delete**
Dashboards	**6**	**Create, add charts, get info, list, update, delete**
Data Discovery	6	Search, schema fields, entity details, dataset queries
Metadata	10	Descriptions, owners, tags, glossary terms, domains
Lineage	1	Upstream/downstream with configurable depth
Reports & Alerts	4	Create, list, get info, delete
Datasets	4	Create, get info, list, delete
Databases	3	List databases, list schemas, list tables
Tags (Superset)	4	Create, add to object, list, delete
Identity & Admin	4	Identity, MCP token, instance info, health check